Name: RISK 1.3 - unequal performance

<Files\\Processed x Highly relevant\\2024-07-16 - RNZ - Government checks for bias in> - § 1 reference coded [17.23% Coverage]

Reference 1 - 17.23% Coverage

A decade ago, facial recognition sparked controversy worldwide over misidentifying black and brown faces much more than white faces, and especially for women - some systems showed a racial or gender bias of four percent. Many algorithms were 10 to 100 times more likely to inaccurately identify a photo that was not of a white face, it was reported in 2019.
But now some systems are down to 0.02 percent inaccuracy, benchmark testing by the US National Institute of Standards and Technology (NIST) shows.
The racist perception persisted, though it was "wrong", said a US thinktank, Lawfare.
"For all the intense press and academic focus on the risk of bias in algorithmic face recognition, it turns out to be a tool that is very good and getting better," it said.
At New York University, researchers generated 13 million synthetic images of faces from six major racial groups - Black, Indian, Asian, Hispanic, Middle Eastern and White - to train three types of facial recognition.
"The result not only boosted overall accuracy compared to models trained on existing imbalanced datasets, but also significantly reduced demographic bias," they reported in April. They have open-sourced their code for others to use.
"Facial recognition technology continues to get better with time, with improved camera technologies, more diverse datasets," said facial recognition bias expert Ifeoma Nwogu at the University at Buffalo.

<Files\\Processed x Highly relevant\\2024-10-13 - RNZ - Governments Identity Check fac> - § 1 reference coded [3.44% Coverage]

Reference 1 - 3.44% Coverage

Facial recognition AI globally has a history of being less accurate for people with darker skin tones, though it has been on the improve.
"What we can say today is that, from that testing, there is no significant bias that's been picked up through independent testing," Waldron said.
A year ago, his developers were still struggling with an "unconfirmed risk relating to racial bias, or the level to which it may exist, in DIA's technology."

<Files\\Processed x Highly relevant\\2024-10-19 - RNZ - Why Google Maps cant pronounce> - § 4 references coded [13.96% Coverage]

Reference 1 - 2.22% Coverage

It is unacceptable that AI tools and global tech companies are still lagging in te reo pronunciation, a Māori tech innovator says.

Reference 2 - 3.83% Coverage

Jones said the quality benchmark for te reo pronunciation is much lower for global tech companies such as Google, but the bigger problem is that no proficient Māori language speakers are involved in the creation of AI tools.

Reference 3 - 2.07% Coverage

Māori language models - such as Te Hiku Media's model - provide an opportunity to set the benchmark much higher, he said.

Reference 4 - 5.83% Coverage

"Providing quality examples of pronunciations like place names... these enhance the language acquisition opportunities for language learners and mispronunciation often obscures the intelligence we hope to convey through speech... What I would say is that accepting a 50 percent accuracy as a benchmark for te reo Māori is just unacceptable."

<Files\\Processed x Highly relevant\\2024-11-08 - SCP - AI Expert Concerned No Maori W> - § 2 references coded [17.74% Coverage]

Reference 1 - 1.97% Coverage

A facial recognition expert is rejecting an official test that cleared a government system of racial bias.

Reference 2 - 15.78% Coverage

While it only had 148 people in it, the test report said "in the context of this evaluation looking at fairness for different demographic groups, the sample size was sufficient to undertake a comparative bias analysis". Of those tested, 61 percent were of European or Asian ethnicity, and 39 percent were Māori or Pasifika.
The system has to be able to catch impostors trying to use it for false verifications. Tests of this returned higher error rates for Māori and Pasifika which the report said might be due to blur, or lighting, or "shadows".
"If there's shade on a face which can create an issue recognising a face, what happens when a person has a moko on their face?" Tairua said. "It's likely to also contribute to false negative matches" - that is, the system treating a genuine selfie as if it were from an impostor. "It's not clinical.

<Files\\Processed x Highly relevant\\2024-11-21 - TAM - Experts explore using AI in> - § 1 reference coded [1.75% Coverage]

Reference 1 - 1.75% Coverage

“We know that big corporates benchmark at 50% for marginalised languages and indigenous languages like te reo Māori.
“50% is the di~erence between a maunga [pronounced wrongly] and a maunga.”

<Files\\Processed x Highly relevant\\2024-12-03 - RNZ - Adopting a single AI app for g> - § 1 reference coded [4.75% Coverage]

Reference 1 - 4.75% Coverage

Internal Affairs has worked with the Ministry of Social Development to use its fledgling Identity Check tool that verifies a person via a selfie put through facial recognition.
Over 70,000 people used it in its first year, and DIA is promoting it further, now it has passed a test for racial bias in a small test on New Zealanders. Concerns among some Māori remain.